

MSC2050 Discrete Mathematics, Week 15

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Spring 2024
Version 1.0 typed under $\text{\LaTeX}$

Some elements of abstract algebra

Groups

A **group** $(G, \cdot)$ is a nonempty set G together with a binary operation $\cdot$ on G such that the following conditions hold:

- (i) Closure: For all $a, b \in G$ the element $a \cdot b$ is a uniquely defined element of G .
- (ii) Associativity: For all $a, b, c \in G$, we have $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- (iii) Identity: There exists an identity element $e \in G$ such that $e \cdot a = a \cdot e = a$ for all $a \in G$.
- (iv) Inverses: For each $a \in G$ there exists an inverse element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

Examples of infinite groups:

$(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are groups.

$(\mathbb{Q} \setminus \{0\}, \cdot)$, $(\mathbb{Q}_+, \cdot)$, $(\mathbb{R} \setminus \{0\}, \cdot)$ are groups.

$(\mathbb{Z}, \cdot)$ is not a group.

The set of all invertible $n \times n$ matrices forms a (non-commutative) group under multiplication.

Examples of finite groups:

$(\{1, -1\}, \cdot)$ is a group.

$(\mathbb{Z}_n, +_n)$ is a group. In particular $(\{0, 1\}, +_2)$ is a group.

For finite groups the number of elements is called the **order of G** .

Cancellation Property for Groups

Let $(G, \cdot)$ be a group, and let $a, b, c \in G$.

(i) If $ab = ac$, then $b = c$.

(ii) If $ac = bc$, then $a = b$.

Outline of the proof: $b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}ac = c$.

$(G', \cdot)$ is a **subgroup** of $(G, \cdot)$ if $G' \subseteq G$ and $(G', \cdot)$ is a group.

Theorem

If $G' \subseteq G$ is finite, then $(G', \cdot)$ is a subgroup if G' is closed under $\cdot$.

Outline of the proof: Clearly associative property will hold for $(G', \cdot)$. Let $a \in G'$. Hence, $a^2, a^3, a^4, \dots \in G'$. Since G' is finite and closed under addition, $a^i = a^j$ for some $i < j$. Hence, $e = a^{j-i} \in G'$ and $a^{-1} = a^{j-i-1} \in G'$.

Let $(G, \cdot)$ be a group $a \in G$ and $H \subseteq G$.

The (left) **coset of H with respect to a** is denoted by $a \cdot H$ is the set

$$a \cdot H = \{a \cdot x \mid x \in H\}.$$

Theorem

Let H be a subgroup of $(G, \cdot)$. Then either $a \cdot H$ and $b \cdot H$ are disjoint or they are identical.

Proof: Suppose $a \cdot H \cap b \cdot H \neq \emptyset$. Then there is $c \in a \cdot H \cap b \cdot H$. Hence, $c = a \cdot h_1 = b \cdot h_2$. Therefore, $a = b \cdot h_2 \cdot h_1^{-1}$.

Let $x \in a \cdot H$. Then $x = a \cdot h_3 = b \cdot h_2 \cdot h_1^{-1} \cdot h_3 \in b \cdot H$. So $a \cdot H \subseteq b \cdot H$. Similarly, $b \cdot H \subseteq a \cdot H$. Hence, $a \cdot H = b \cdot H$.

Answer Q1

Example

Let $(B^6, \oplus)$ be a group of bit strings of length 6 with bitwise operation $\oplus$ defined by:

$$a_1 a_2 \dots a_6 \oplus b_1 b_2 \dots b_6 = a_1 +_2 b_1 \ a_2 +_2 b_2 \dots a_6 +_2 b_6.$$

Let

$$a^0 = 000000 \quad a^1 = 001111 \quad a^2 = 010011 \quad a^3 = 011100$$

$$a^4 = 100110 \quad a^5 = 101001 \quad a^6 = 110101 \quad a^7 = 111010$$

$(\{a^0, a^2, \dots, a^7\}, \oplus)$ is a subgroup of $(B^6, \oplus)$.

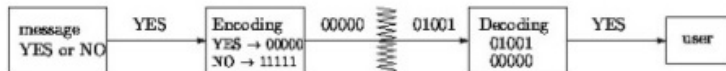
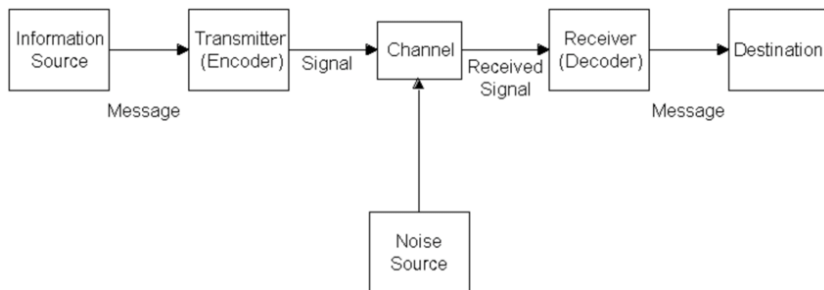
The following table gives the cosets of $(B^6, \oplus)$.

000000	001111	010011	011100	100110	101001	110101	111010
100000	101111	110011	111100	000110	001001	010101	011010
010000	011111	000011	001100	110110	111001	100101	101010
001000	000111	011011	010100	101110	100001	111101	110010
000100	001011	010111	011000	100010	101101	110001	111110
000010	001101	010001	011110	100100	101011	110111	111000
000001	001110	010010	011101	100111	101000	110100	111011
110000	111111	100011	101100	010110	011001	000101	001010

Coding theory

(Paragraph 12.6 from the hard copy)

Cryptography develops methods to protect information against an enemy.
Coding theory develops methods to protect information against a noise.
Coding theory allows not only to **find errors** but also **correct errors** in the codes.
Coding theory methods are often elegant applications of very basic concepts and methods of **abstract algebra**.



Main idea

Examples of channels: telephone lines, atmosphere...

Examples of noise: sunspots, meteor showers, radio disturbance, poor typing, poor hearing...

Method of fighting errors: **REDUNDANCY!**

The key idea is that in order to protect a message against a noise, we should encode the message by adding some **redundant information** to the message.

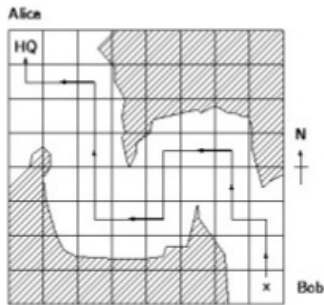
In this case, even if the message is corrupted by a noise, there will be enough redundancy in the encoded message to recover, or to decode the message completely.

For example, In encoding $0 \rightarrow 000$ and $1 \rightarrow 111$

Then in decoding we have $000, 001, 010, 100 \rightarrow 0$ and $111, 110, 101, 011 \rightarrow 1$.

Alice and Bob example

Alice has a map and only she knows the route through which Bob can reach her avoiding the enemy territory. Alice wants to send Bob the information about the safe route



NNWNNWWSSWWNNNNWWN

1. $N \rightarrow 00$, $W \rightarrow 01$, $E \rightarrow 10$ and $S \rightarrow 11$.

Then any error in 00000100000101111010100000000010100 would be a **disaster!**

2. $N \rightarrow 000$, $W \rightarrow 011$, $E \rightarrow 101$ and $S \rightarrow 110$.

A single error in each symbol could be **detected** (even number of 1's).

3. $N \rightarrow 00000$, $W \rightarrow 01101$, $E \rightarrow 10110$ and $S \rightarrow 11011$.

A single error in each symbol could be **corrected** (repeat first two numbers at the end and add 0 in the middle if they are the same and 1 if they are different).

Two errors can be detected.

Let B^n be the set of all binary strings of length n .

An (m, n) -code for a binary message consists of an encoding scheme $E : B^m \rightarrow B^n$ and a decoding scheme $D : B^n \rightarrow B^m$ ($m < n$), which tries to make the transmission of m bit binary word error-free.

Consider an encoding scheme $e : B^m \rightarrow B^{5m}$.

$$e(a_1 \dots a_m) = a_1 \dots a_m a_1 \dots a_m a_1 \dots a_m a_1 \dots a_m a_1 \dots a_m$$

If $m = 2$, $e(10) = 1010101010$.

The corresponding decoding scheme is defined as follows:

$$d(b_1 \dots b_{5m}) = a_1 \dots a_m,$$

$a_i = 1$ if 3 or more of $b_i, b_{i+m}, b_{i+2m}, b_{i+3m}, b_{i+4m}$ are 1's and 0 otherwise.

Suppose, while transmitting 1010101010, the message received is 1011101010 with an error. Then $d(1011101010) = 10$.

Double errors also can be detected $d(1011011010) = 10$.

Answer Q2

Hamming distance

The **weight** of $x \in B^n$ denoted $w(x)$ is the number of 1's in x .

The **Hamming distance** between $x, y \in B^n$ denoted $d(x, y)$ is the weight of $x \oplus y$ (excluding OR (XOR)).

For example, $x = 011001$ and $y = 101000$.

Then $w(x) = 3$, $w(y) = 2$, $x \oplus y = 110001$, $d(x, y) = 3$.

The Hamming distance satisfies the following properties:

- $d(x, y) = d(y, x)$
- $d(x, y) \geq 0$
- $d(x, y) = 0$ if and only if $x = y$
- $d(x, y) \leq d(x, z) + d(z, y)$. (For proof: $w(x \oplus y) \leq w(x) + w(y)$ and $x \oplus y = x \oplus (z \oplus z) \oplus y = (x \oplus z) \oplus (z \oplus y)$)

The **minimal distance** of a code is

$$M := \min\{d(E(m_1), E(m_2)) \mid m_1, m_2 \in B^m, m_1 \neq m_2\}$$

In the Alice and Bob example to code the safe route

NNWNNWWSSWWNNNNWWN

1. $N \rightarrow 00$, $W \rightarrow 01$, $E \rightarrow 10$ and $S \rightarrow 11$. $M = 1$.

Then any error in 000001000001011111010100000000010100 would be a **disaster!**

2. $N \rightarrow 000$, $W \rightarrow 011$, $E \rightarrow 101$ and $S \rightarrow 110$. $M = 2$.

A single error in each symbol could be **detected**.

3. $N \rightarrow 00000$, $W \rightarrow 01101$, $E \rightarrow 10110$ and $S \rightarrow 11011$. $M = 3$.

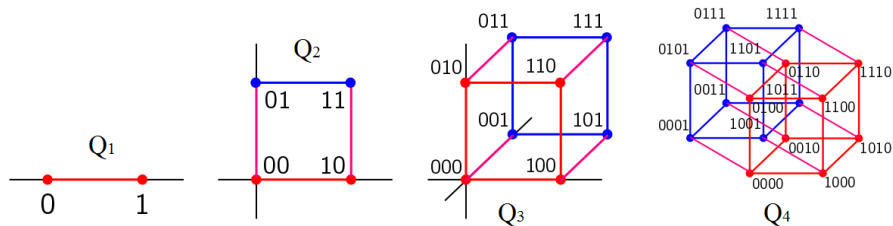
A single error in each symbol could be **corrected**.

Two errors can be detected.

M	1	2	3		4		5			6		
C	0	0	0	1	0	1	0	1	2	0	1	2
D	0	1	2	1	3	2	4	3	2	5	4	3

n -dimensional cube Q_n

The n -dimensional cube can be built recursively and drawn as a graph.



n -dimensional cube has vertices: 2^n .

Every vertex has degree n .

Hence n -dimensional cube has edges: $2^{n-1}n$.

$d(a, b)$ is the length of a minimal path from a to b .

Theorem

Up to $M - 1$ errors can be detected.

Minimum distance decoding criterion

Suppose a codeword x was sent and a word y was received. We can correct (decode) y if $d(y, x) < d(y, x')$ for all codewords x' .

Basic error correcting theorem

If $M = 2t + 1$, then up to t errors can be corrected, when the minimum distance decoding criteria is considered.

proof: Assume there were t errors and $2t + 1$ is the minimal distance of the code $e : B^m \rightarrow B^n$.

Suppose a codeword x was sent and a word y was received. Then $d(x, y) \leq t$.

Let x' be another codeword. Then $d(x, x') \geq 2t + 1$.

Since

$$d(x, x') \leq d(x, y) + d(y, x'),$$

it follows that $d(y, x') \geq t + 1$.

Answer Q3

Conclusion

If we know that channel makes not more than t errors then we choose (n, m) -code so that $M = 2t + 1$.

How can we can choose? **Abstract algebra** helps here.

Group codes

A subset G of B^n is called a **group code** if $(G, \oplus)$ is a subgroup of $(B^n, \oplus)$.

Theorem

The minimum distance of the group code is the minimum weight of a non-zero codeword.

Proof: Let $M = d(x, y)$ and $w = |z|$ be a minimum weight for $x, y, z \in G$. We have that $x \oplus y \in G$ since G is a subgroup and so

$$M = d(x, y) = w(x \oplus y) \geq w.$$

On the other hand, $w = |z| = w(x \oplus 0) = d(z, 0) \geq M$. Hence, $w = M$.

Let

$$a^0 = 000000 \quad a^1 = 001111 \quad a^2 = 010011 \quad a^3 = 011100$$

$$a^4 = 100110 \quad a^5 = 101001 \quad a^6 = 110101 \quad a^7 = 111010$$

$$M = 3$$

Boolean matrices in coding scheme

A coding scheme $e : B^m \rightarrow B^n$ ($m < n$) can be constructed using **Boolean matrix** E of size $m \times n$. The first $m \times m$ submatrix forms an identity matrix. The coding words are given by $e(x) = x * E$.

Theorem

The set of codewords defined by a Boolean matrix as above form a group.

Proof: Let $x', x'' \in B^m$ and $x = x' \oplus x''$, so $e(x) = e(x' \oplus x'')$.

Let $e(x) = y_1 y_2 \dots y_n$ and $E = \{e_{ij}\}$ so

$$y_j = \sum_{i=1}^m x_i e_{ij} = \sum_{i=1}^m (x'_i \oplus x''_i) e_{ij}.$$

We also have that

$$\sum_{i=1}^m (x'_i \oplus x''_i) e_{ij} = \sum_{i=1}^m x'_i e_{ij} \oplus \sum_{i=1}^m x''_i e_{ij}.$$

Hence, $e(x) = e(x') \oplus e(x'')$. It follows that

$$y = e(x) = e(x' \oplus x'') = e(x') \oplus e(x'') = y' \oplus y''.$$

The closure is proved. So the set of codewords form a group.

Decoding

Let $e : B^2 \rightarrow B^5$ be given by the matrix

$$I_2 A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

So the encoding gives

$$e(00) = 00000 \quad e(01) = 01101 \quad e(10) = 01101 \quad e(11) = 11110$$

The matrix

$$H = A^T I_3 = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

is called **parity check matrix**. If $Hc^T = 0$, then c is a codeword. If $Hc^T \neq 0$, then error has occurred. If $c = 10111$ has received then

$$Hc^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

which is the same as column 3 of H , which means that the third bit is an error and corrected is 10011. If Hc^T is not 0 or any column of H , more than one error has occurred, so cannot be corrected ($M=3$).

Answer Q4

Decoding in Group codes using cosets

- Determine all cosets of G .
- For each coset pick the word with the smallest weight (leader of a coset).
- For received y , $x \oplus y$ is the sent word, where x is the leader of the coset containing y

Let

$$\begin{aligned} a^0 &= 000000 & a^1 &= 001111 & a^2 &= 010011 & a^3 &= 011100 \\ a^4 &= 100110 & a^5 &= 101001 & a^6 &= 110101 & a^7 &= 111010 \end{aligned}$$

The following table gives the cosets of $(B^6, \oplus)$.

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000100	001011	010111	011000	100010	101101	110001	111110
000010	001101	010001	011110	100100	101011	110111	111000
000001	001110	010010	011101	100111	101000	110100	111011
110000	111111	100011	101100	010110	011001	000101	001010

For received 011110, the codeword is $011110 \oplus 000010 = 011100$.

In the last row double error has occurred ($M=3$).

Answer Q5

Hamming codes

Perfect codes which will correct **all single errors** but **cannot detect double errors** is given by R.W.Hamming.

- Choose an integer r . The code is $(2^r - 1 - r, 2^r - 1)$
- Form a matrix H of $2^r - 1$ rows and r columns, where the row i is the binary number.
- For $r = 3$, $a_1a_2a_3a_4 \rightarrow b_1b_2a_1b_4a_2a_3a_4$ ($b_{2^r-1} = a_{2^r-1-r}$)
The bits b_1, b_2, b_4 are found from $bH = 0$.
- For received y , if $yH = 0$ then no errors. If yH is the i th row of H then the error has occurred in the i th bit of y .

Consider $(4, 7)$ Hamming code ($r = 3$). Received $b = 0001011$.

$$bM = 0001011 \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = 101$$

This shows the fifth bit is in error, so corrected is 0001111.

Homework:

- Solve odd exercises starting at p. 826.